

EXPERIMENT 16: PERFORM EDGE DETECTION USING CANNY METHOD

Aim

To detect edges in an image using the Canny Edge Detection algorithm in Python and OpenCV.

Algorithm

1. Import OpenCV library.
2. Read the input image.
3. Convert the image into grayscale.
4. Apply the Canny edge detector.
5. Display and save the output image.

Python Code

```
import cv2

# Read image
img = cv2.imread("input.jpg")

if img is None:
    print("Error: Image not found")
    exit()

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Apply Canny Edge Detection
edges = cv2.Canny(gray, 100, 200)

# Save output
cv2.imwrite("canny_output.jpg", edges)

# Display images
cv2.imshow("Original Image", img)
cv2.imshow("Canny Edge Detection", edges)
```

```
cv2.waitKey(0)  
cv2.destroyAllWindows()
```

Input



Output



Result

Thus, the edges in the image were successfully detected using the Canny Edge Detection algorithm.